GF(16) cubic GHS extension: exact class-C quartic witnesses

26 October 2026. Conductor T-5361. The frozen protocol precedes the draw. The screen, raw cohort, independent replay, replay certificate, and prior-work ledger preserve the exact run.

New finding and claim boundary

The GF(16)/GF(4096) extension found three exact class-C-only source orders with a large selected prime subgroup and a smooth plane-quartic Jacobian in the same finite-field isogeny class. The equations are new **entries in this research screen** and are absent from the LMFDB `curves` arrays queried on 26 October 2026; those arrays currently list only hyperelliptic models for all eleven matched classes. This is a database/model gap, not proof of publication novelty or a new vulnerable family. An explicit source-to-quartic map and its nonzero action on the prime subgroup remain unconstructed.

The screen drew 2,048 quartics. The filter tallies were `N1=17`: 231, `N2=257`: 14, smooth: 12, full Weil-polynomial/source-order matches: 12. Those matches covered eleven distinct source orders. The source cohort contains all 8,190 normalized ordinary elliptic models, with both absolute trace classes of `a` and every nonzero `b`.

Exact curves and subgroups

Let `K=GF(2)[w]/(w^12+w^7+w^6+w^5+w^3+w+1)` and `k=GF(2)[a0]/(a0^4+a0+1)`, using Sage's polynomial-basis integer encoding. Each source has equation `E(a,b): y²+xy=x³+a*x²+b`. Quartic coefficient vectors follow monomials `x^i*y^j*z^(4-i-j)` in descending `i=4..0`, then descending `j=4-i..0`. The zero leading coefficient is forbidden; all listed vectors begin with the `x⁴` coefficient one.

#E(K)	a,b	selected r, cofactor	fixed G, Q=[7]G	#C(K)	LMFDB class	fully split distinct rational lines
4036	0,112	1009,4	(4030,3567), (2128,2753)	3914	3.16.a_a_acj	5
4124	0,34	1031,4	(2705,1890), (23,1463)	4178	3.16.a_a_bb	9
4126	32,63	2063,2	(62,1499), (2633,1798)	4184	3.16.a_a_bd	7

The exact plane quartic vectors are:

- Order 4036: `[1,9,13,10,14,7,15,10,13,7,15,14,2,5,0]`.
- Order 4124: `[1,3,12,3,0,0,2,13,13,6,5,8,0,10,10]`.
- Order 4126: `[1,13,13,0,10,7,11,10,5,10,14,5,15,3,11]`.

For every selected source, `[r]G=[r]Q=0`. The independently reconstructed quartics are geometrically smooth, have `N1=17` and `N2=257`, and their Jacobians have the displayed source orders. Rational point-difference classes reach each selected order-`r` subgroup. These checks give a factor-base entry point, not an elliptic-to-Jacobian homomorphism. The five, nine and seven fully split four-point lines cannot alone span a 17-column relation system; other relation types would be necessary.

Independent full-cohort audit

The verifier uses a point-count identity independent of Sage's elliptic cardinality routine. For `x≠0`, set `u=1/x²` and `z=y/x`. The equation becomes `z²+z=x+a+b*u`; absolute trace satisfies `Tr(x)=Tr(x²)=Tr(u⁻¹)`. With `q=4096`, therefore,

`#E(a,b)(K) = q+1 + (-1)^Tr(a) Σ_{u≠0} (-1)^Tr(u⁻¹+b*u).`

The verifier computes all `b` sums as a 4096-entry Walsh transform after mapping each `u` to the trace-dual coordinate vector `(Tr(u*w^0),...,Tr(u*w^11))`. Its 8,190 resulting orders exactly match the screen population. It recounts each retained quartic by rooting in the alternate projective charts `(u:1:v)`, `(1:0:v)`, `(0:0:1)`, rather than the screen's `(1:u:v)` charts. This replay checked all eleven retained point-count triples.

Prior work for every matched class

The exact full-polynomial query to LMFDB found all eleven classes. Listed model counts equal hyperelliptic counts in every row; the three highlighted labels link to their class records: 4036, 4124, and 4126.

Source order	LMFDB label	stored models / hyperelliptic	class-B source present?
3978	`3.16.a_a_aep`	20 / 20	no
4008	`3.16.a_a_adl`	76 / 76	yes
4030	`3.16.a_a_acp`	56 / 56	no
4036	`3.16.a_a_acj`	156 / 156	no
4094	`3.16.a_a_ad`	128 / 128	no
4098	`3.16.a_a_b`	104 / 104	yes
4106	`3.16.a_a_j`	100 / 100	yes
4108	`3.16.a_a_l`	140 / 140	no
4124	`3.16.a_a_bb`	64 / 64	no
4126	`3.16.a_a_bd`	140 / 140	no
4140	`3.16.a_a_br`	100 / 100	no

Order 4106 is a direct class-B positive control with a `2053` prime subgroup and cofactor two, so it belongs to the prior-known direct binary GHS family. Exact-label web searches for the three highlighted classes returned no source identifying these particular quartic equations; that bounded search cannot establish they were unpublished.

Transfer-method audit

Tian's cubic isogeny construction provides a known quotient idea. Its displayed Weil functions and quartic reconstruction use a full symplectic geometric two-torsion basis, three nonzero two-torsion points on a short Weierstrass equation, and a classical bitangent Aronhold system. Ordinary characteristic-two elliptic curves have one nonzero geometric two-torsion point. Consequently, those **published coordinate steps cannot be directly run on the three binary sources**. An odd-degree quotient adapted to characteristic two remains a research possibility. This methodological gap does not make the overall attack method new; Joux–Vitse and Hess's binary Weil descent already supply related cover and descent routes.

Attack-cost gate and novelty labels

Order	new exact curve	new eligible family	new attack method	evaluated subgroup map	full GHS cost vs rho
4036	unresolved outside the catalog	no	no	missing	unknown
4124	unresolved outside the catalog	no	no	missing	unknown
4126	unresolved outside the catalog	no	no	missing	unknown

An eligible vulnerability claim needs an evaluated map `Phi` with `Phi(G)≠0`, `Phi(Q)=[7]Phi(G)`, a complete relation/decomposition and target-descent solve, independently replayed scalar, and total same-target cost below rho. The current results establish none of those attack-cost conditions. The existing checked Lean subgroup theorem remains conditional on constructing `Phi`; it is not instantiated by these structural matches.

Reproduction

```
sage -python research/ghs_binary_cubic_isogeny_q16_20261026/screen.py
sage -python research/ghs_binary_cubic_isogeny_q16_20261026/verify.py
```

The raw result binds the protocol and screen SHA-256; the replay certificate binds the raw result and verifier SHA-256. Elapsed screen time is diagnostic and is not an attack cost.